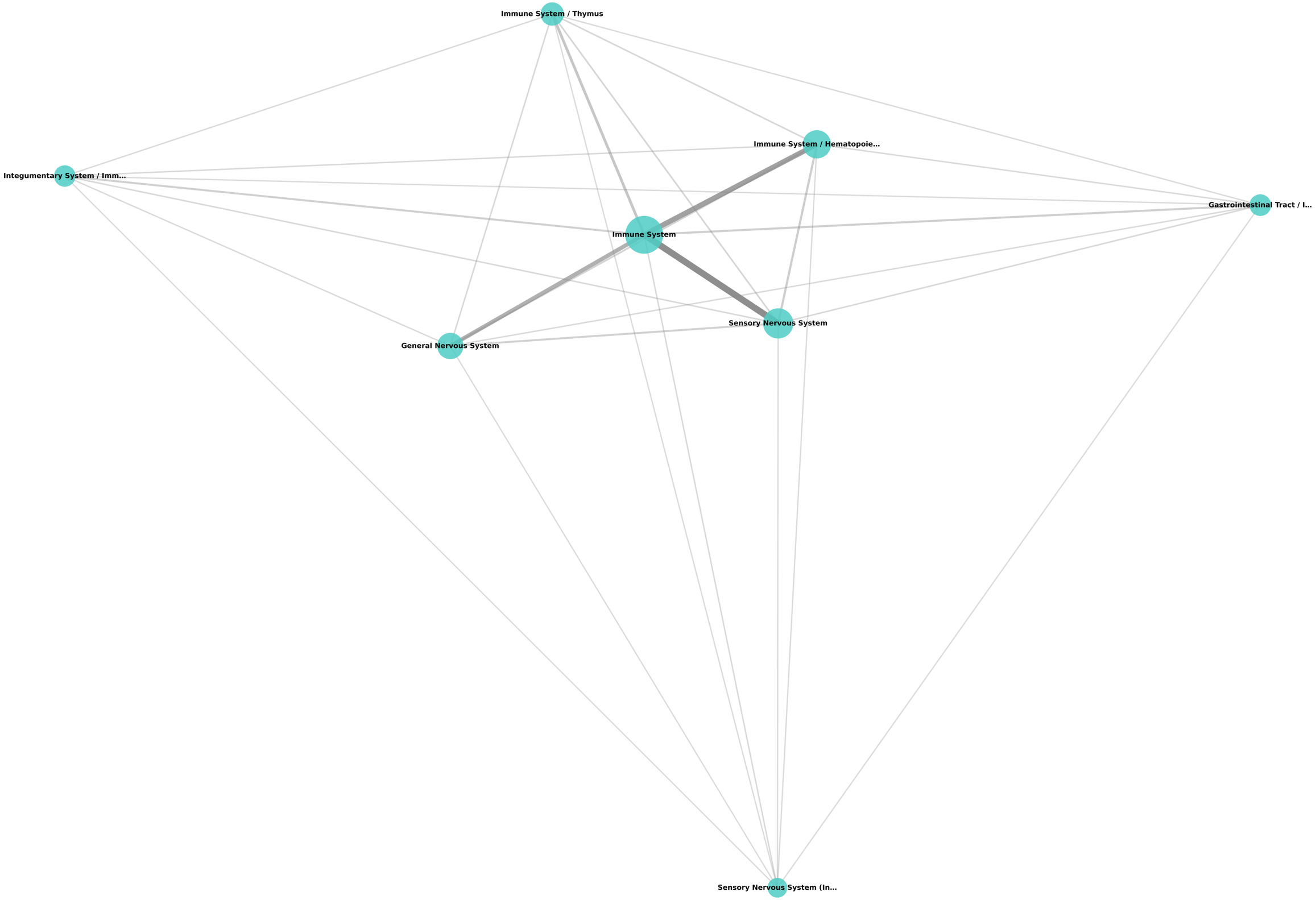Broad group connectivity — pathway edges